Active buzzer

DO | RASPBERRY PI | SOUND

What does it do?

An active buzzer makes one built-in tone. Switch it on and off for warning beeps or rhythms; its pitch stays the same.

Recognise it: check the packet for **active** or **built-in oscillator**. Shape, stickers and + marks do not prove the type.

Use a labelled **3.3 V GPIO-compatible** buzzer.

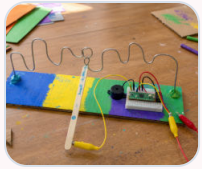
Uses



Smoke alarm

One warning tone is enough; no changing notes needed.

DETECT Measure smoke level → **DECIDE** Is smoke above the limit? → **DO** Repeat one warning tone



Buzz wire game

Choose active: one fixed buzz clearly signals failure.

DETECT Loop touches wire → **DECIDE** Is this the third strike? → **DO** Buzz at one pitch



+ leg
Marked +, to GPIO

− leg
Other leg, to GND

Vent
Sound comes out

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import Buzzer

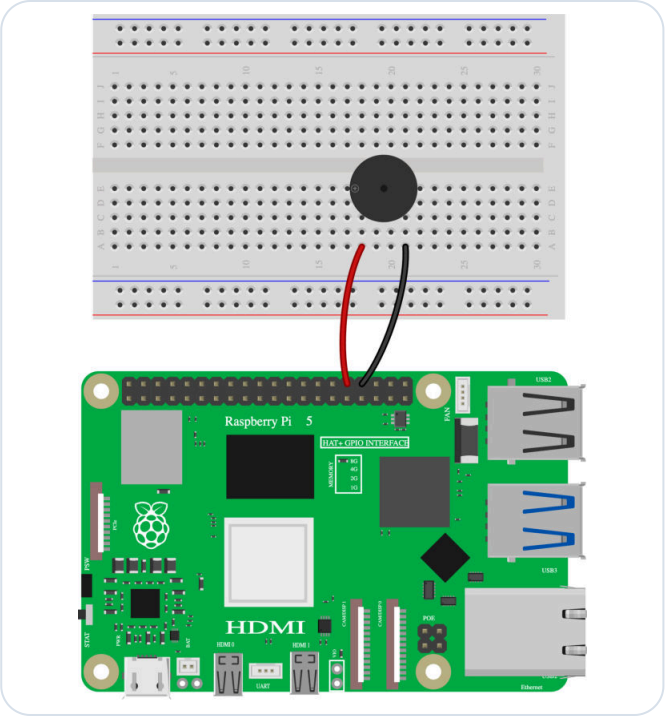
buzzer = Buzzer(12)

buzzer.on()
```

Also `buzzer.off()`, `buzzer.toggle()` and `buzzer.beep()`. Give `beep()` an `on_time` and an `off_time` to set the rhythm. Faster beeps do not mean a higher pitch.

Wiring

RASPBERRY PI 5



GPIO 12 (pin 32) → **+** buzzer **−** → **GND** (pin 34)